

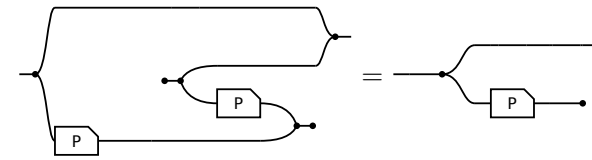
Relation Algebra

§5. Domain and range

(5a)

The **domain** $\text{dom}(R) \triangleq \mathbb{1} \multimap (R \multimap R^\circ)$ and the **range** $\text{ran}(R) \triangleq \text{dom}(R^\circ)$.

(5b)



$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\multimap$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{ran}(R)$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

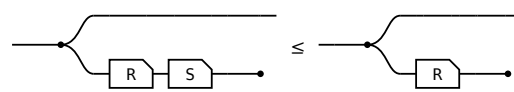
(5c)

$\text{dom}(R) \triangleq \mathbb{1} \multimap (R \multimap R^\circ)$
$\text{Coreflexive}(X) \implies \text{dom}(R) \sqsubseteq X \iff R \sqsubseteq X R$
$\text{dom}(R S) \sqsubseteq \text{dom}(R)$
$\text{dom}(R \multimap S) = \mathbb{1} \multimap (S \multimap R^\circ)$
$R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = \mathbb{1}$
$R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ R \sqsubseteq \mathbb{1}$
$R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$
$\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(R S)$ $\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(f g)$ $\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(f g)$
$\text{Entire}(R S) \implies \text{Entire}(R)$

§5.1. Sliding the discard

$\text{dom}(RS) \sqsubseteq \text{dom}(R)$, and a single glyph for **dom** would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



$S \multimap \multimap$, the lax axiom for $\multimap$ in the first section
— the discard slides back past S

Equality is S entire, which is the same picture read as $\text{dom}(R) = \mathbb{1} \iff R \text{ entire}$.